

3 (a) Some plant cell walls develop secondary thickening.

Diagram 1 shows a cell without secondary thickening.

Diagram 2 shows part of the cell walls of two adjacent cells with secondary thickening.

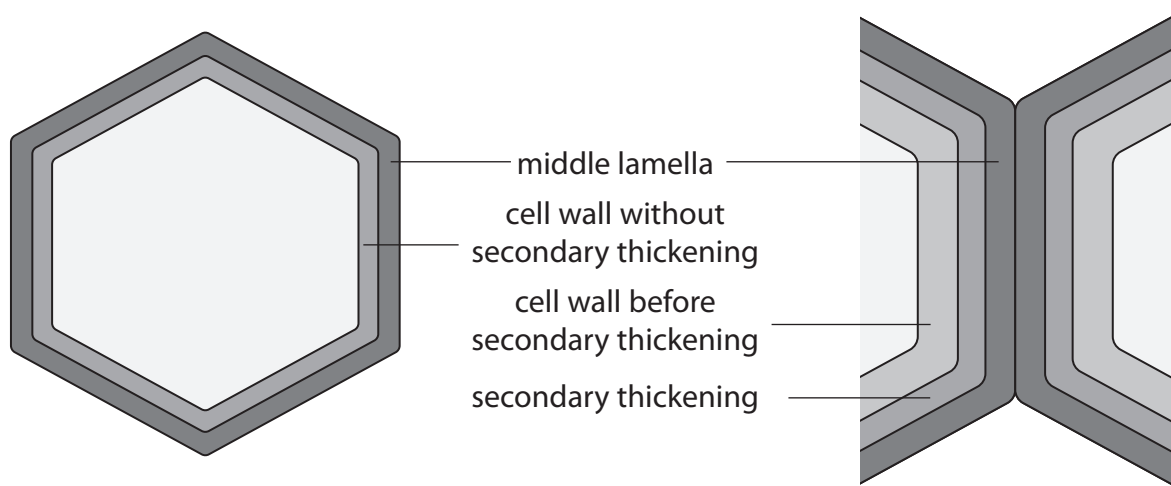


Diagram 1

Diagram 2

Give **three** differences between a cell wall with secondary thickening and a cell wall without secondary thickening.

Use the information in the diagrams to support your answer.

(3)

- 1
- 2
- 3

(b) (i) Plant cell walls contain cellulose.

How many of the following statements about cellulose molecules are correct?

- contain α -glucose molecules
- contain 1,4 and 1,6 glycosidic bonds
- form microfibrils

(1)

☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three

(ii) Secondary thickening occurs in sclerenchyma fibres.

Explain how secondary thickening contributes to the physical properties of sclerenchyma fibres.

(2)

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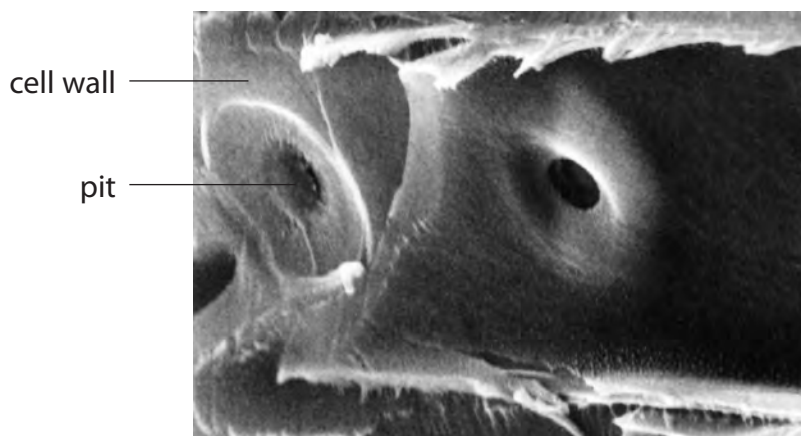
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(c) Some cells in xylem vessels have secondary thickening.

The photograph shows pits in the wall of part of a xylem vessel, as seen using an electron microscope.



(Source: G.F. Gennaro / Science Photo Library)

Describe the function of pits in the xylem vessel.

(2)

(Total for Question 3 = 8 marks)